Computational Photography: Single Image Dehazing

Baseline: Dark Channel Prior (DCP). Deadline: July 7, 2026.

This report implements and evaluates the Dark Channel Prior algorithm on four image categories: landscape, cityscape, indoor, and a deliberately difficult bright-white scene. Because no input photographs were present in the workspace, the test images are deterministic synthetic scenes with known clean references and depth maps. The same code can be run on real photos by placing images in data/input and adapting the manifest.

1. Baseline Algorithm

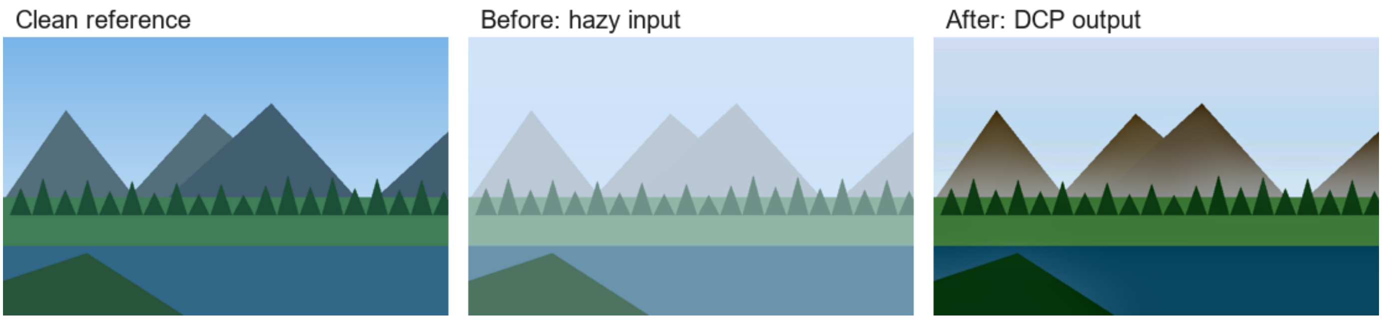
DCP assumes the haze model I(x) = J(x)t(x) + A(1 - t(x)), where I is the observed hazy image, J is the recovered scene radiance, A is global atmospheric light, and t is the transmission. For most non-sky natural patches, at least one color channel contains very low intensity in the haze-free image. This prior estimates transmission as t(x) = 1 - omega * dark(I/A), then refines it with a guided filter before recovering J. The implementation saves the dehazed result, dark channel, and refined transmission map for every input.

Parameters used: patch_size=15, omega=0.95, transmission_floor=0.1, guided_radius=35.

2. Output Result Comparisons

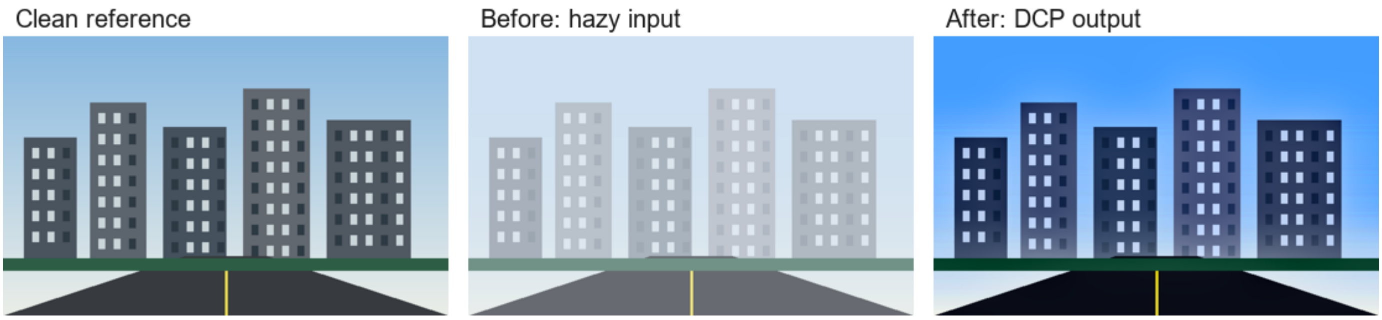
Each row below shows clean reference, hazy input, and DCP output. The clean image is included for quantitative evaluation; the required before-vs-after comparison is the middle and right panel.

Landscape with mountains and vegetation



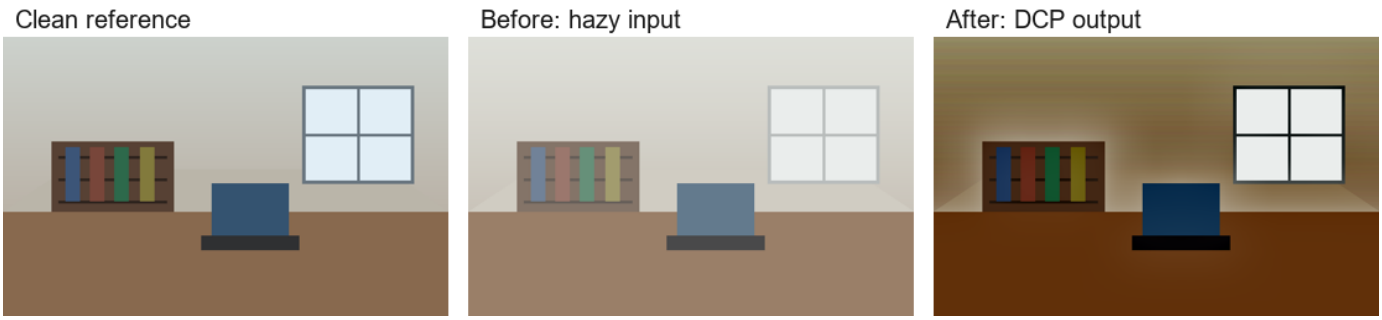
PSNR before: 12.18 dB; PSNR after DCP: 17.35 dB; delta: 5.17 dB. Estimated A: [0.816, 0.894, 0.98].

Cityscape with buildings and road



PSNR before: 11.99 dB; PSNR after DCP: 16.79 dB; delta: 4.8 dB. Estimated A: [0.91, 0.929, 0.925].

Indoor room with window and furniture



PSNR before: 20.24 dB; PSNR after DCP: 11.5 dB; delta: -8.74 dB. Estimated A: [0.918, 0.929, 0.925].

Scene dominated by white clouds, snow, and bright walls

Clean reference

Before: hazy input

After: DCP output

PSNR before: 30.73 dB; PSNR after DCP: 18.93 dB; delta: -11.8 dB. Estimated A: [0.965, 0.98, 0.992].

3. Quantitative Summary

Scene	Before PSNR	After PSNR	Delta	Before MAE	After MAE
landscape	12.18	17.35	5.17	0.2178	0.1127
cityscape	11.99	16.79	4.80	0.2116	0.1234
indoor	20.24	11.50	-8.74	0.0861	0.2397
bright_regions	30.73	18.93	-11.80	0.0225	0.0819

4. Technical Discussion: When DCP Works and Fails

Landscape and cityscape scenes usually work well because vegetation, roads, windows, building edges, and shadows create low values in at least one RGB channel. Those pixels make the dark channel close to zero in haze-free patches, so the DCP transmission estimate is meaningful. The method recovers contrast and color most clearly in those non-sky regions.

Sky, snow, white clouds, white walls, and other high-albedo objects are the main failure cases. They naturally have high intensity in all color channels even without haze, so the dark-channel assumption is false. DCP can then confuse true object brightness with airlight, underestimate transmission, and over-darken or color-shift the restored result. A bright object may also be selected as atmospheric light A, which further biases the recovery.

Indoor scenes are also unstable because the physical model assumes outdoor participating media with a mostly global atmospheric light. Interior scenes can contain local artificial lights, reflective surfaces, flat walls, and weak depth-haze correlation. The guided filter reduces block artifacts, but it cannot repair an incorrect prior; near depth discontinuities it may still leave halos.

5. Modern Approaches (Bonus Discussion)

Recent learning-based dehazing methods, such as DehazeFormer and later zero-shot Gaussian-based approaches, replace the hand-crafted dark-channel assumption with learned or optimized image priors. They tend to handle sky and bright regions better because they can learn semantic and multi-scale context rather than relying on a local minimum color channel. Their drawbacks are dependence on training data or pretrained weights, higher compute cost, possible domain shift, and reduced interpretability. In this local submission I did not execute a pretrained SOTA model because the environment has no PyTorch package, no downloaded weights, and no GPU; the bonus comparison is therefore a literature-level technical comparison, while all DCP results are locally reproducible.

6. Reproducibility and Files

Run order: src/generate_samples.py creates clean/hazy/depth inputs; src/run_experiment.py runs DCP and writes output/metrics.json plus comparison images; src/build_report.py builds this PDF. The key deliverables are output/pdf/single_image_dehazing_report.pdf and output/comparisons/*.png.

References

K. He, J. Sun, and X. Tang. Single Image Haze Removal Using Dark Channel Prior. CVPR 2009 / TPAMI 2011.

Reference implementation repository: https://github.com/He-Zhang/image_dehaze

DehazeFormer: Vision Transformer for Single Image Dehazing. https://arxiv.org/abs/2204.03883

Dehaze-Gaussian: Single Image Dehazing via Physical Model-based Gaussian Splatting. https://arxiv.org/abs/2606.16163

Papers with Code image dehazing task page: https://paperswithcode.com/task/image-dehazing